

Data Analytics Track

1. Purpose

Part E validates the most important analytical findings from the final FinTrust solution. Each finding is checked using an independent validation method and classified according to whether the evidence supports, revises or partially supports the original conclusion. The validation draws on SQL, Python and Power BI outputs produced during the project.

Validation structure: Finding → Validation Method → Result → Conclusion

2. Final Validation Results

1. Monthly transaction performance

Finding	Transaction activity declined in February 2026 and recovered strongly in March.
Validation Method	Compared the final SQL monthly trend query with Python monthly aggregation and the Power BI monthly trend visual.
Result	All three tools showed the same pattern: January 4,133 transactions / NGN 188.49M; February 3,734 / NGN 175.79M; March 4,133 / NGN 196.20M.
Conclusion	Supported. The February decline and March recovery are consistent across SQL, Python and Power BI.

2. Risk review increases with transaction value

Finding	Higher-value transactions are more likely to be selected for risk review.
Validation Method	Compared SQL amount-band analysis, Python amount-band recalculation and the Power BI risk-review-by-amount-band visual.
Result	The tools matched across all amount bands. The risk-review rate rose from 17.66% below NGN 10K to 40.54% for NGN 200K and above.
Conclusion	Supported. Transaction value is strongly associated with risk-review selection, especially above NGN 200K.

3. Failure hotspots vary by channel and transaction type

Finding	Transaction failures are concentrated in specific Channel × Transaction Type combinations.
Validation Method	Compared the SQL failure-rate query with the Python heatmap and the Power BI failure-rate matrix.
Result	POS + Bill Payment remained the highest observed failure-rate combination at 7.22%, followed by USSD + Transfer at 7.20%.
Conclusion	Supported. Failure analysis should be performed at the combined channel and transaction-type level rather than only overall.

4. Segment totals can hide per-customer performance

Finding	Overall segment totals do not fully describe customer-level performance.
Validation Method	Compared the normalized SQL segment analysis with Power BI measures for transaction value per customer and transactions per customer.
Result	SME customers generated the highest transaction value per customer (NGN 387,923.02), while Student customers recorded the highest transaction frequency (8.23 transactions per customer).
Conclusion	Supported. Normalized KPIs reveal differences that are hidden when only total segment activity is considered.

3. Final Validation Summary

The final validation confirmed strong consistency across the key FinTrust analytical findings. Monthly performance, transaction failure patterns, risk-review behaviour and normalized segment performance were reproduced across the relevant analytical tools. Overall, the validation process supports the reliability of the final business insights and demonstrates that the conclusions presented in the final dashboard and documentation are evidence-based.